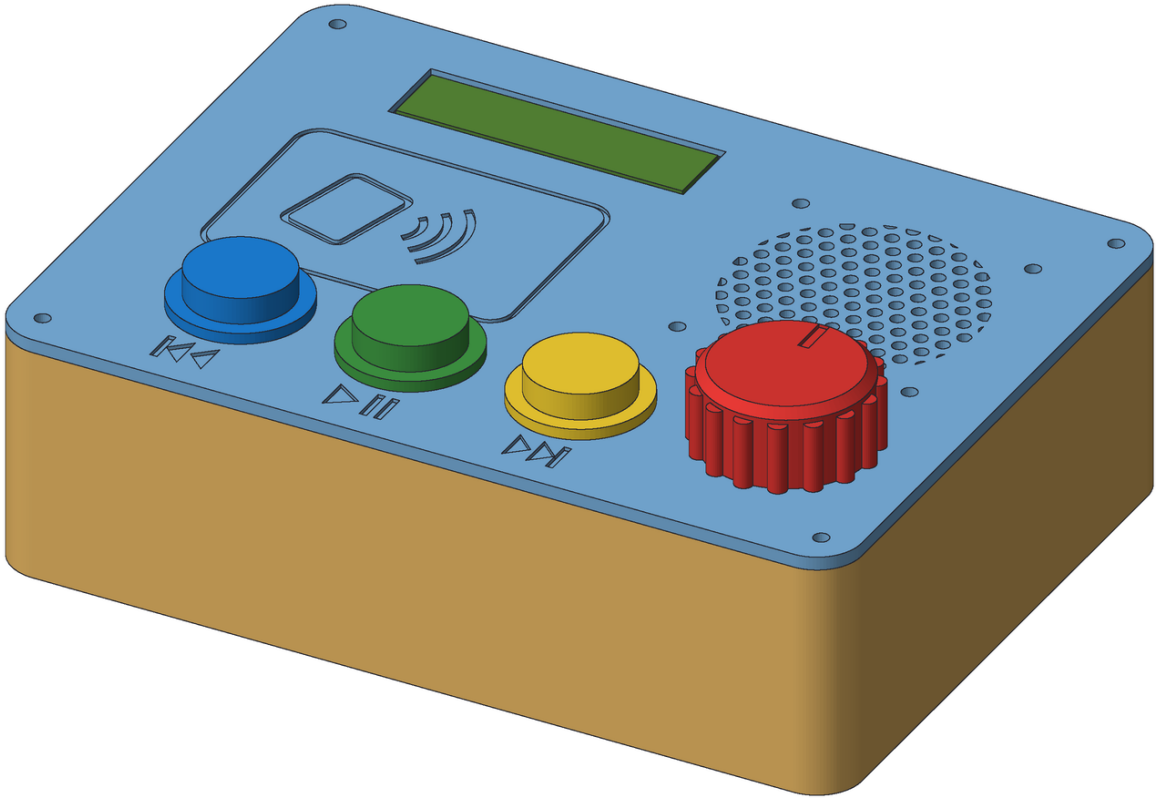
RFID Music Box: build guide

Phase 3. From breadboard to a box a 4-year-old can use: parts, 3D printing, wiring, assembly.



What you are building

• A printed box, 190 x 140 x 55 mm, in two parts: a **base** holding the Uno and a **lid** with everything the child touches.
• On the lid: the LCD, a **tap here** zone over the card reader, three big arcade buttons (previous, play/pause, next), a large volume knob, and the grille for a 2.5-inch speaker with its own small amplifier.
• The wiring moves to a **proto shield** that plugs onto the Uno. Every lid part connects to it with a latching plug, so the lid lifts off for repairs.
• Power and new cards: the Uno's USB port is a hole in the left wall. A USB cable with an on/off switch powers it, and you can still upload new cards without opening the box.

Who does what

• **Your friend:** prints 4 files (page 3). About 300 g of PETG and 12-15 hours of printing.
• **You:** order the parts (page 2), solder the shield (pages 4-5), assemble (pages 6-7).

Before you start

• The code now expects the **I2C LCD** and the buttons on **D2, D3, D4**. Once the new LCD arrives, wire it on the breadboard and test everything there first (page 4 has the pins).
• Tools: soldering iron and solder, side cutters, hot glue gun, the hex key from the screw kit, a small screwdriver.
• Files: `assets/3d/` in the repo (STL files plus `make_box.py`, which makes them).

Safety: nothing loose inside can be reached without a screwdriver. Hot glue stays inside the box. At full volume the speaker is loud up close: if it is too loud, lower `MAX_VOLUME` in `src/main.cpp`.

Shopping list

Amazon US links. Prices checked on 16 October 2026 and they change often. Most packs leave spares.

#	Part	What it's for	Pack	Price	Amazon
1	HiLetgo UNO R3 Proto Shield	The permanent circuit board: plugs onto the Uno, everything is soldered to it. Peel off the little breadboard and solder to the pads.	2	$7.79	B00HHYBWPO
2	GeeekPi I2C LCD1602, blue	Same screen with an I2C backpack already soldered on: 4 wires instead of 12, with a contrast screw on the back. Address 0x27.	2	$9.99	B07S7PJYM6
3	EG STARTS 24 mm arcade buttons	The three buttons. Snap-in, with microswitches inside. Six colours, two of each.	12	$12.99	B01N5JRU2R
4	Guwarry 2.8 mm button cables	Push onto a button's switch tabs at one end and plug into the shield (JST-XH) at the other. No soldering at the buttons.	20	$9.99	B07RRSG321
5	TWTADE B10K potentiometer	Panel-mount volume knob. Comes with a JST-XH 3-pin cable and a small knob that the printed knob fits over.	3	$9.99	B082FCRQS2
6	XH 2.54 mm connector kit	Pins for the shield, plus pre-crimped cables for the LCD, reader and speaker. Includes heat-shrink tubing.	kit	$15.99	B08G18PWQ6
7	HiLetgo female header strips	A socket for the DFPlayer, so it plugs in and can be swapped.	10	$5.99	B00VVI1L1W
8	ruthex M3 heat-set inserts	Brass threads melted into the base, so the lid screws can come out and go back in many times.	100	$9.99	B08BCRZZS3
9	MCMASKE M2/M2.5/M3 screw kit	Uses M3 x 8 mm (lid, Uno, speaker), M3 nuts (speaker), M2.5 x 6 mm (LCD) and the washers.	kit	$14.99	B0B5KWJVCY
10	ANDTOBO USB on/off switch cable	Goes between the charger and the Uno's USB cable. The switch turns the box on and off.	1	$6.99	B07T9BRNHW
11	Dayton Audio PC68-4 2.5" speaker	A real full-range driver, 4 ohm, much fuller sound than a tiny module speaker. Mounts under the lid's round grille.	1	$11.89	B0751FW9NP
12	HiLetgo PAM8403 amp boards	Small 5 V class-D amp for the speaker. It takes the DFPlayer's line output (DAC pins) instead of the DFPlayer's weak built-in amp.	5	$5.69	B00LODGV64
	Total			**$122.28**	

Optional

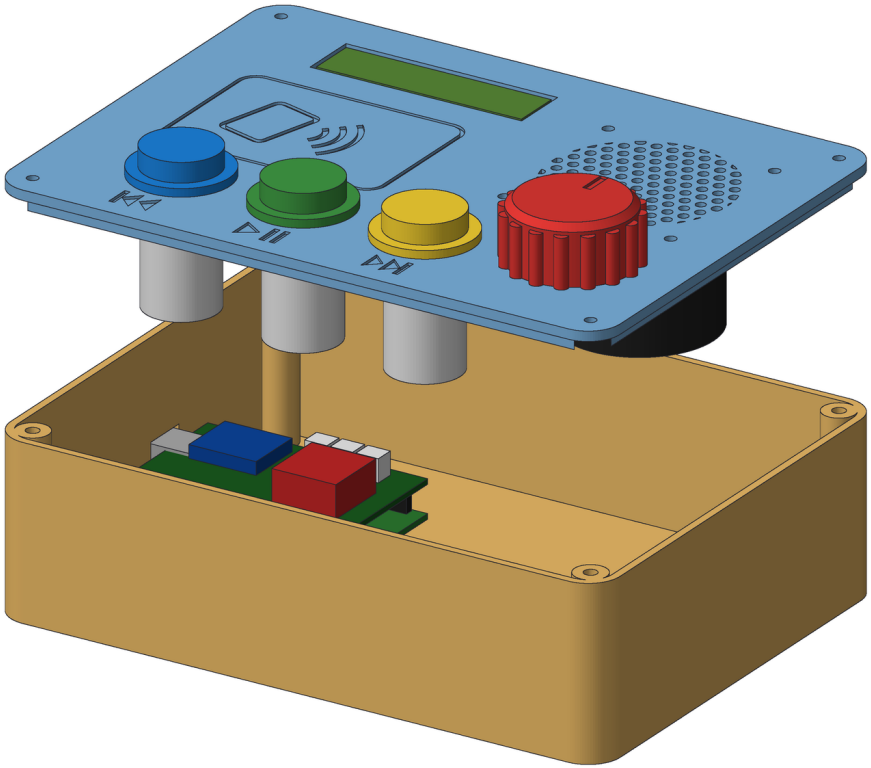
Part	When you need it	Price	Amazon
NTAG213 NFC stickers (20)	To make figurines: your friend pauses a print halfway, you drop a sticker in, the print finishes over it. The reader sees them like cards.	$5.95	B0DQV8NJ5Z
Heat-set insert kit with 60 W soldering iron	Only if you don't own a soldering iron. It replaces item 8.	$27.99	B0D2KB8743
TUOFENG 22 AWG solid wire, 6 colours	Only if you have no hook-up wire for the short jumpers on the shield.	$16.19	B07TX6BX47
Stick-on rubber feet, 10 mm	Any brand. The base has four shallow round pads for them.		

Already on hand

Arduino Uno and its USB cable, RC522 reader, DFPlayer Mini, microSD card with the music, the 1k ohm resistor, two 10k ohm resistors and a 100 uF capacitor (both in the Elegoo kit), a 5 V / 2 A USB charger, the 13.56 MHz cards. The old LCD, its 5.1k and 220 ohm resistors and the CQRobot speakers are no longer needed. Your friend supplies the PETG filament (about 300 g).

For the 3D printer

Four STL files in assets/3d/. All are already turned the right way up for printing, and none needs supports.



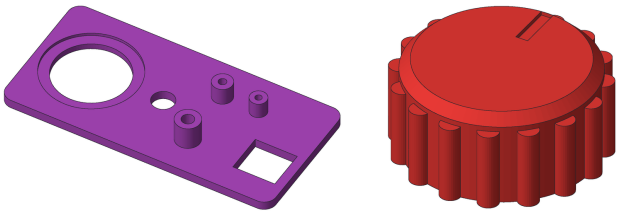
File	What it is	Print notes
fit_test.stl	Small plate with one of each critical hole	**Print this first** (about 30 min). Checks below.
base.stl	The bottom tray, 190 x 140 x 52 mm	Open side up. The longest print (about 6-9 h).
lid.stl	The top, with posts and rims underneath	Already **upside down**: the top face goes on the bed, which makes it smooth and keeps the icons crisp. About 3-4 h.
knob.stl	Big volume knob, 36 mm	Opening down. Glue it over the pot's own small knob.

Settings

- **PETG**: it takes drops better than PLA. PLA works too, but keep the box out of a hot car.
- 0.2 mm layers, **4 walls**, 4 top and bottom layers, 20% infill (gyroid or grid).
- Two colours look good: the base in one, the lid in another.
- Any bed of 200 x 150 mm or more fits the base and lid.
- The speaker sits in a square rim under the lid, sized for a 68.1 mm frame. Measure the speaker when it arrives; if it differs, change `SPEAKER_FRAME` before printing the lid.

Fit test: check before printing the big parts

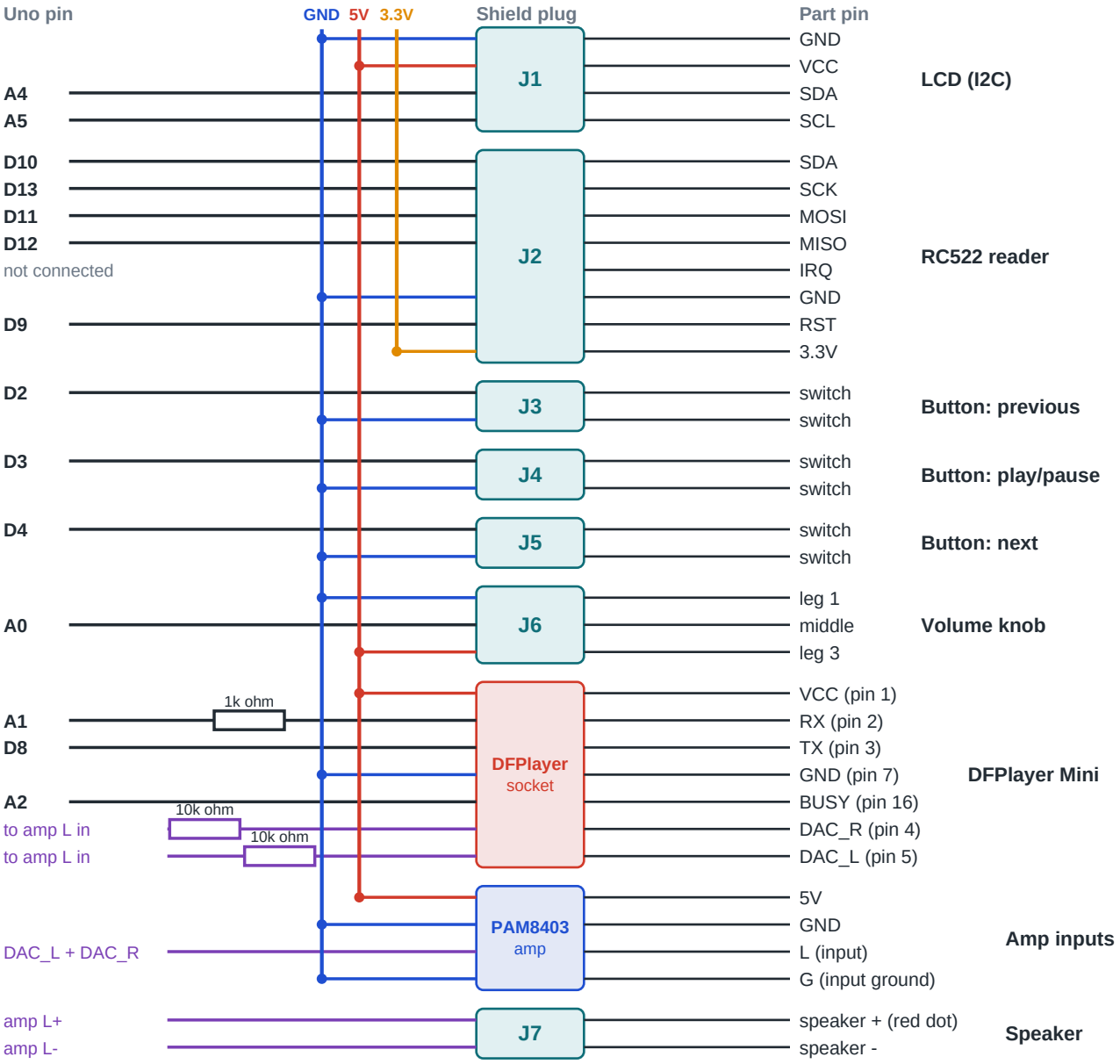
- **Big hole:** an arcade button snaps in and holds firmly. Loose: lower `BUTTON_HOLE_D`. Won't snap: lower `BUTTON_PANEL_T`.
- **Small hole:** the potentiometer thread passes through and its nut goes on.
- **Thick post:** a heat-set insert melts in straight. **Medium post:** an M3 screw cuts its own thread. **Thin post:** an M2.5 screw does the same.
- **Square hole:** the Uno's USB cable plug goes in square, without forcing.
- Adjust the numbers at the top of `make_box.py`, run `python3 make_box.py` (it needs `pip install manifold3d trimesh numpy`), and print again.



fit_test and knob

Wiring: what goes where

Each wire as it runs from the Uno pin, through the shield, to a plug, to the part. The colours match the GND, 5V and 3.3V rails.



Uno pins

Pin	Goes to
D2, D3, D4	Buttons: previous, play/pause, next (each to GND when pressed)
D8	DFPlayer TX
D9-D13	RC522 (RST, SDA, MOSI, MISO, SCK)
A0	Knob middle leg
A1	1k ohm, then DFPlayer RX
A2	DFPlayer BUSY
A4, A5	LCD SDA, SCL (I2C)
3.3V	RC522 only. **Never 5V.**
5V, GND	LCD, DFPlayer, knob; GND to everything
Free	D5, D6, D7, A3. D0/D1 stay free for USB.

Changed from phase 2: the LCD is now I2C on A4/A5 (it used D2-D7 and two resistors), and the buttons moved from A3-A5 to D2-D4. The speaker now runs from a PAM8403 amp fed by the DFPlayer's DAC pins; the DFPlayer's SPK pins are left unconnected.

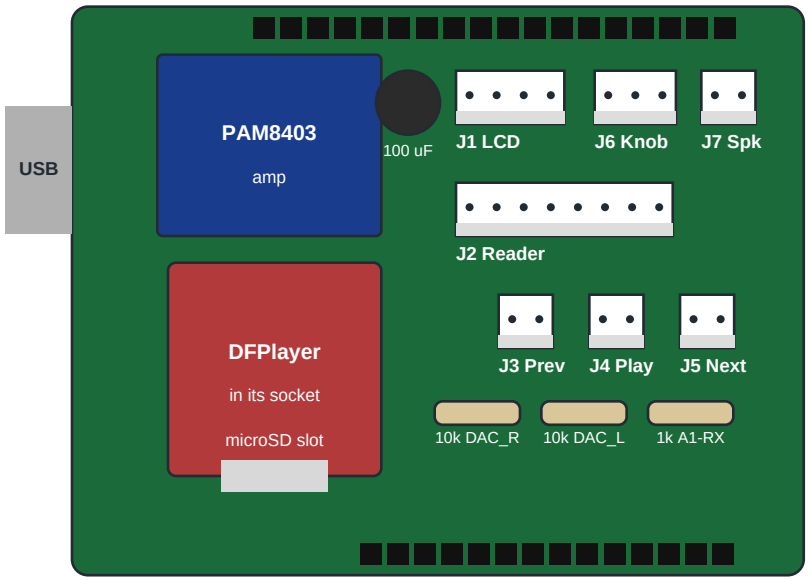
Amp outputs: L+ and L- both go to the speaker. **Never** connect L- (or R-) to GND. The R input and outputs stay empty.

LCD contrast: turn the blue screw on the backpack until the letters are sharp. If the screen stays blank, the Serial Monitor says `LCD not found`: check its 4 wires.

Building the shield

A suggested layout, seen from the top with the USB jack on the left (the way it sits in the box). Solder on the underside; route jumpers on top.

back of the box (LCD side): digital header D0-D13



front of the box (button side): power and analog header

Grey strip = the plug's locking side. Pin 1 of each plug is its left end.

Steps

1. Solder the shield's stacking headers if they aren't on already. Peel off the mini breadboard; the pads under it are what you solder to.
2. **DFPlayer socket:** cut two 8-pin pieces of female header and solder them so the DFPlayer plugs in, close to the front edge, with its microSD slot facing that edge. Then the card comes out when the lid is off. Mark pin 1 (VCC) on the board with a pen.
3. Solder the **1k ohm resistor** from A1 to the DFPlayer's RX pin.
4. **Amp:** solder the PAM8403 flat near the back-left corner, on short wires or header pins. Its 5V and GND go to the rails, with the **100 uF capacitor** across them (the stripe side to GND). A **10k resistor** from DAC_R (pin 4) and another from DAC_L (pin 5) meet at the amp's **L** input; the amp's input **G** goes to GND. Its **L+** and **L-** outputs go to J7. The DFPlayer's SPK pins stay empty.
5. Solder the **XH pins** J1-J7 (from the connector kit) where they're easy to reach, with the locking side facing out. Write the J number next to each.
6. Solder short jumpers between pads following page 4. Make a **GND** line and a **5V** line along the board first, then tap every GND and 5V pin into them. 3.3V goes **only** to J2 pin 8.
7. Check with a multimeter before plugging anything in: 5V to GND must **not** beep. 3.3V to 5V must not beep. L- to GND must not beep.

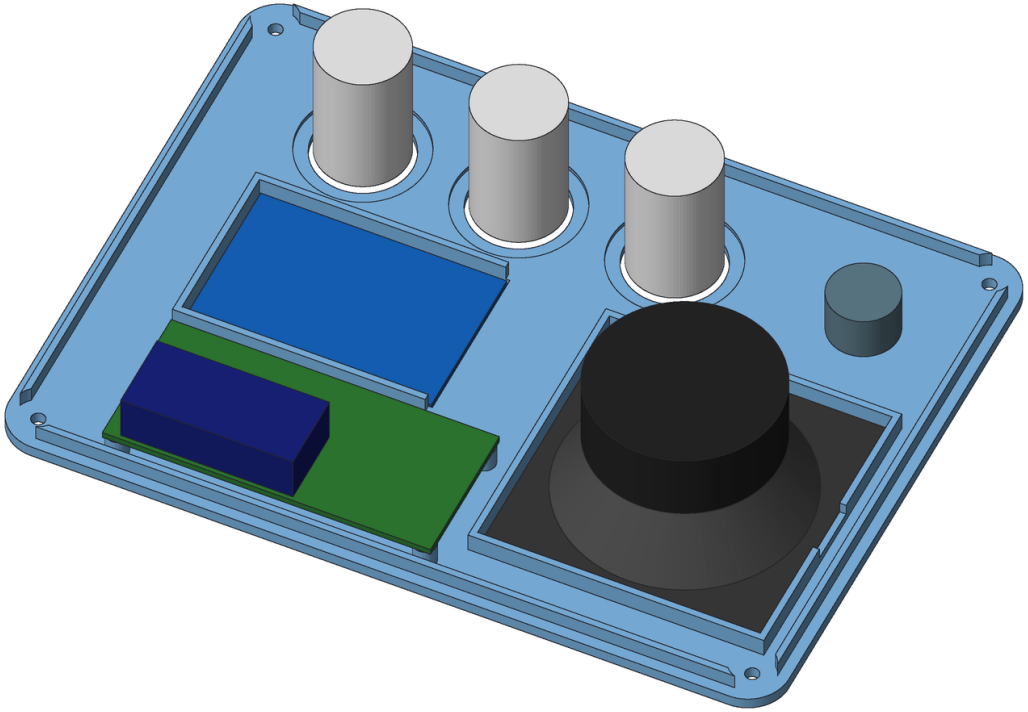
Cables for the lid parts

- **LCD (J1):** a 4-wire XH cable from the kit. Solder the loose ends to the backpack's pins GND, VCC, SDA, SCL, in that order. Cover each joint with heat-shrink.
- **Reader (J2):** an 8-wire XH cable, soldered to the RC522's 8 pins in order, SDA to 3.3V. Pin 5 (IRQ) stays unconnected at both ends.
- **Buttons (J3-J5):** one Guwarry cable each. The flat clips push onto the switch's two tabs, either way round. If a switch has three tabs, use COM and NO.
- **Knob (J6):** comes with its cable. If turning clockwise makes it quieter, pop the outer two wires out of the plug (lift the tab with the tweezers) and swap them.
- **Speaker (J7):** solder a 2-wire XH cable to the speaker's two tabs: J7 pin 1 (amp L+) to the tab marked with a red dot or +. Heat-shrink both joints.

Leave about 15 cm of cable on each lid part, so the lid can rest beside the box with everything still plugged in.

Assembly 1: the lid

Lay the lid upside down on a towel. Everything below mounts from underneath, except the buttons and the knob.



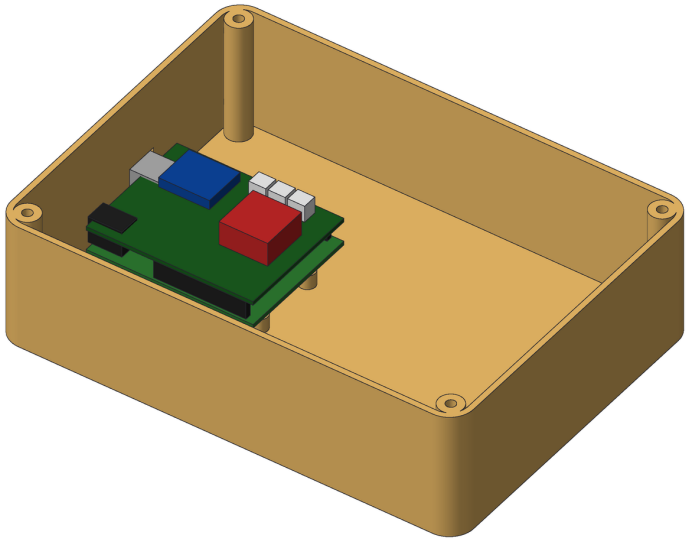
Steps

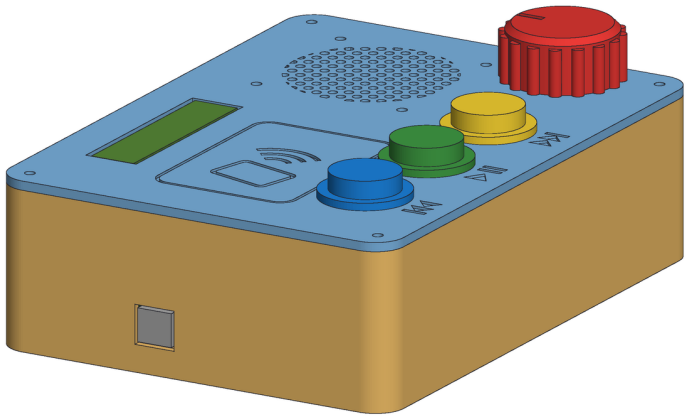
1. **Buttons:** push each one in from the top until its clips snap, previous on the left (|<< icon), play/pause in the middle, next on the right. Pick bright colours; the pack has six.
2. **Volume knob** (front right): take the nut and washer off the potentiometer. Push its shaft up through the small hole from underneath, then washer and nut on top. Snap off the small anti-turn tab on the pot if it stops it sitting flat. Push the pot's own knob on, then glue the printed knob over it (super glue or epoxy).
3. **LCD:** the metal frame drops into the pocket round the window, with the pins toward the back edge. Four M2.5 x 6 mm screws go into the posts. If it rattles, put a washer between each post and the board.
4. **Reader:** lay the RC522 in its pocket inside the low rim, with the pins toward the open side. A dot of hot glue on each corner holds it. Keep screws and the speaker magnet away from it; metal shortens the reading range.
5. **Speaker:** cone face down into its square rim, over the round grille, with its tabs toward the gap in the rim. Four M3 x 8 mm screws go in from the top of the lid through the frame's corner holes, with an M3 nut on the back of each. Snug, not tight.
6. Plug nothing in yet. Check that the lid's corner holes line up with the base.

Screws into plastic: the posts have no thread. The screw cuts its own the first time, so turn it slowly and stop when it's snug. Over-tightening strips the hole.

Assembly 2: the base, first power-on, and closing up

The base holds the Uno, the shield and the DFPlayer. The USB jack lines up with the hole in the left wall.





Steps

1. **Inserts:** heat the iron to about 230 °C (PETG). Set an insert on top of each corner pillar and let the iron press it straight down until it's flush. Let it cool.
2. **Uno:** tilt it so the USB jack enters the wall hole from inside, then lower it onto the four posts. Fix it with M3 x 8 mm screws. If the post next to the headers is too tight for the screwdriver, three screws are enough.
3. Plug the **shield** onto the Uno. Push the microSD card into the DFPlayer, then the DFPlayer into its socket, pin 1 on your mark.
4. Stand the lid beside the box and plug in J1-J7.
5. **First power-on with the lid open**, using the checklist on the right.
6. **Close:** tuck the cables in, lower the lid so its lip drops inside the walls, and fix the four corners with M3 x 8 mm screws. Stick the rubber feet into the round pads.

First power-on

• Charger, then the switch cable, then the Uno's USB cable into the wall hole. Switch on.
• LCD: `Starting...`, then `Tap a card...`. Blank or blocks: turn the contrast screw.
• Tap a card on the **tap here** mark: its folder plays, and the LCD shows the title and `Song 1`.
• Each button: previous, play/pause, next.
• Knob: clockwise is louder.

If something is wrong

• `RFID not found`: check J2, and that pin 8 is on 3.3V.
• `MP3 not found`: SD card in, DFPlayer the right way round, 1k resistor in place.
• Card not seen: the reader sits flat in its pocket, nothing metal near it, and the card is 13.56 MHz.
• A button does nothing: its cable is on the right J, and the clips are fully on the tabs.
• No sound but `Song 1` shows: check the amp's 5V, the two 10k resistors, and J7.
• Distorted when loud: lower `MAX_VOLUME` (try 25). Hum: check the 100 uF capacitor and the amp's input G to GND.
• Reboots when loud: use a 2 A charger, or lower `MAX_VOLUME`.

Everyday use

• **New card:** tap it, read the ID, add it to `src/cards.h`, and upload with a computer plugged into the wall hole. No need to open the box.
• **New music:** take out the four lid screws, lift the lid, and pull the microSD card out of the DFPlayer.